

Q1: Find the period of $f(t) = (10\cos t)^2$, using identity:-

$$\cos x \cdot \cos y = 0.5(\cos(x-y) + \cos(x+y))$$

Q2: A signal has a fundamental frequency of 1000 Hz.

What is the period in Nano seconds??

Q3: Express the following in their simplest form:

$$\sin(2\pi ft) + \sin(2\pi ft - \pi)$$